

Policy questions in the Solow model

- ▶ What saving rate maximizes steady-state consumption?
- ▶ How does an increase in saving affect consumption along the transition?
- ▶ Which forms of foreign aid can change income temporarily or permanently?

We use $\hat{y} = \hat{k}^\alpha$, $\hat{c} = (1 - s)\hat{y}$, and $\dot{\hat{k}} = s\hat{k}^\alpha - (\delta + n + x)\hat{k}$, with $0 < \alpha < 1$.

The Golden Rule: steady-state consumption

In a steady state, $\hat{i}^* = (\delta + n + x)\hat{k}^*$, so

$$\hat{c}^* = \hat{y}^* - \hat{i}^* = (\hat{k}^*)^\alpha - (\delta + n + x)\hat{k}^*.$$

Maximizing steady-state consumption over $\hat{k}^*$ gives

$$\frac{d\hat{c}^*}{d\hat{k}^*} = \alpha(\hat{k}^*)^{\alpha-1} - (\delta + n + x) = 0.$$

Thus

$$\hat{k}_{\text{GR}} = \left(\frac{\alpha}{\delta + n + x} \right)^{1/(1-\alpha)}.$$

Which saving rate implements the Golden Rule?

Recall the steady-state capital stock as a function of the saving rate:

$$\hat{k}^*(s) = \left(\frac{s}{\delta + n + x} \right)^{1/(1-\alpha)}.$$

Compare this with the Golden Rule capital stock:

$$\hat{k}_{GR} = \left(\frac{\alpha}{\delta + n + x} \right)^{1/(1-\alpha)}.$$

The two expressions have the same form. Therefore,

$$s_{GR} = \alpha.$$

Interpretation

Under Cobb–Douglas production, the saving rate that maximizes steady-state consumption equals capital's income share.

Saving above or below the Golden Rule

$s > \alpha$: overaccumulation

$\hat{k}^* > \hat{k}_{GR}$. Lowering s raises consumption immediately and in the new steady state. The initial capital stock is **Dynamically Inefficient**.

$s < \alpha$: underaccumulation

$\hat{k}^* < \hat{k}_{GR}$. Raising s lowers consumption initially but raises it eventually. Whether the change improves welfare depends on how future consumption is valued.

$$\hat{c}^*(\hat{k}) = \hat{k}^\alpha - (\delta + n + x)\hat{k} \text{ reaches its maximum at } \hat{k}_{GR}.$$

A higher saving rate: transition

At date t_0 , s rises permanently; $\hat{k}$ is predetermined.

1. Investment $s\hat{y}$ rises at once; $\dot{\hat{k}}$ becomes positive.
2. $\hat{k}$ and $\hat{y}$ rise gradually toward higher steady-state levels.
3. Growth of Y/L temporarily exceeds x , then returns to x .
4. $\hat{c} = (1 - s)\hat{y}$ falls on impact; its eventual level may rise or fall depending on the initial and new saving rates.

Key distinction

A higher saving rate raises the *level* of output per worker, but does not permanently raise its growth rate in the Solow model.

Foreign aid: consumption and capital goods

Consumption goods

A temporary transfer raises consumption while it lasts. If it leaves saving, capital, and productivity unchanged, it leaves the economy's production path unchanged.

Capital goods or investment finance

If aid finances productive investment when $\hat{k}$ is below its steady state, it speeds capital accumulation during the transfer. After temporary aid ends, capital converges back toward the original steady state unless the transfer changes domestic saving or productivity persistently.

How a temporary investment transfer works

Let a_t be aid per effective worker, with fraction ϕ invested directly. If the remaining share is treated as income and saved at rate s ,

$$\dot{\hat{k}}_t = s\hat{k}_t^\alpha + \underbrace{[\phi + s(1 - \phi)]a_t}_{\text{investment financed by aid}} - (\delta + n + x)\hat{k}_t.$$

For cash aid entirely treated as disposable income, $\phi = 0$ and the extra investment is sa_t ; for in-kind investment aid, $\phi = 1$ and it is a_t . If $a_t = 0$ after a finite period, the capital accumulation equation returns to its original form. The lasting effect on $\hat{k}$ vanishes as the economy approaches its original steady state.

Technical assistance and the level of technology

If assistance permanently raises labour-augmenting productivity A_t , output per worker $Y_t/L_t = A_t f(K_t/(A_t L_t))$ can be permanently higher. At the instant A_t rises, physical capital K_t is fixed, so $\hat{k}_t = K_t/(A_t L_t)$ falls. Capital then accumulates toward the original steady-state value of $\hat{k}$ (if s, n, x, δ are unchanged).

Level versus growth

A permanent one-time increase in the *level* of A_t raises the long-run level of output per worker. A permanently higher growth rate requires a persistent increase in the growth rate x itself.